Graph Neural Networks Midterm Exam (Fall 2026)

Full Name: _____ Student ID: _____

Note: Time: 120 minutes. 11 problems, 100 points in total. A calculator is allowed; phones and other electronic devices are not. Students should write in English for evaluation. Descriptive answers are graded by keywords, so explain as fully as you can. Show your calculation steps.

1. **(Message Passing)** (10p) Given an undirected graph with the adjacency matrix A, the initial node feature matrix $H^{(0)}$ and the weight matrix W as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \qquad H^{(0)} = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \\ -1 & 2 \end{bmatrix}, \qquad W = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

Assume that one GCN layer is computed as

$$H^{(k+1)} = \sigma\big(\hat{A}\, H^{(k)}\, W\big), \qquad \hat{A} = \tilde{D}^{-1/2}\tilde{A}\,\tilde{D}^{-1/2}, \qquad \tilde{A} = A + I,$$

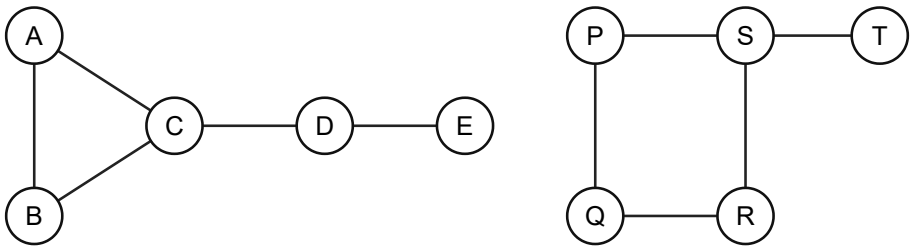
where $\tilde{D}$ is the degree matrix of $\tilde{A}$ ($\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$), I is the identity matrix, and σ is a ReLU function, $\mathrm{ReLU}(x) = \max(0, x)$.

   a. Calculate the row-normalized adjacency matrix $D^{-1}A$, where $D_{ii} = \sum_j A_{ij}$, and the GCN-normalized adjacency matrix $\hat{A}$.

   b. Calculate the output of the GCN layer, $H^{(1)}$.

   c. The node indices in A are arbitrary. Explain why a GNN layer still gives the same output vector to each node when the nodes are re-indexed (permutation equivariance). State this property using a permutation matrix P.

2. **(WL Relabeling, GIN)** (10p) Consider the two undirected graphs G_1 and G_2 given in the following figure.

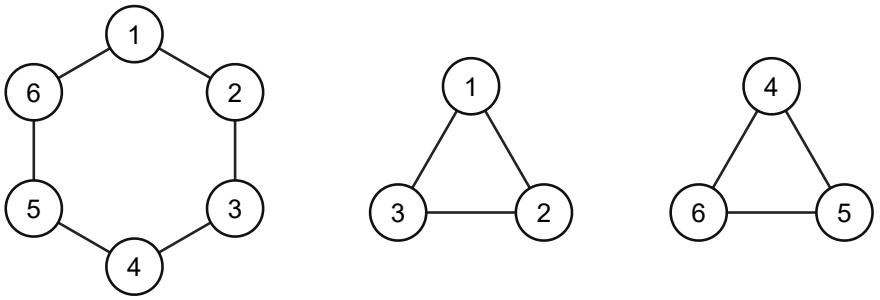


G_1 G_2

   a. Apply the 1-WL (Weisfeiler-Lehman) relabeling to G_1 and G_2. All nodes start with color 1. In each iteration, every node v gets the new color $\mathrm{HASH}\big(c(v), \{\{c(u) : u \in N(v)\}\}\big)$, and both graphs share one HASH table. Show the colors of all nodes after iterations 1 and 2 and the color histograms. Can 1-WL distinguish G_1 and G_2?

   b. GIN updates node features as $h_v^{(k)} = \mathrm{MLP}^{(k)}\big((1+\epsilon^{(k)})\, h_v^{(k-1)} + \sum_{u \in N(v)} h_u^{(k-1)}\big)$. Why is GIN more expressive than ordinary GNNs such as GCN (mean aggregation) or GraphSAGE (max aggregation)?

3. **(Beyond 1-WL, ESAN)** (8p) The graphs G_3 (a 6-cycle) and G_4 (two disjoint triangles) in the following figure cannot be distinguished by 1-WL, since every node has degree 2.

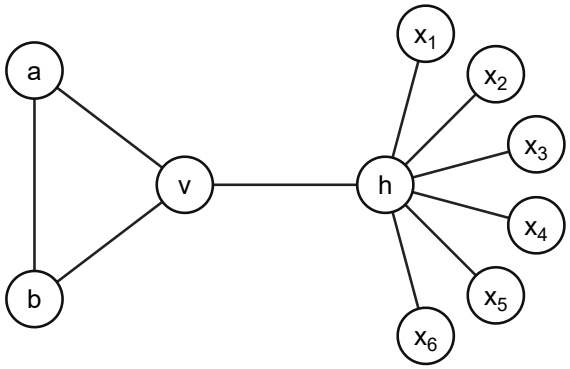


G_3 G_4

a. Explain how ESAN represents a graph as a bag of subgraphs and how it processes the bag. Using the node-deletion policy, explain why ESAN can distinguish G_3 and G_4.

b. Discuss the approaches to make GNNs more expressive than 1-WL: (i) injecting substructure counts (e.g., GSN), (ii) subgraph GNNs, i.e., bag of subgraphs (e.g., ESAN), and (iii) higher-order GNNs (e.g., k-GNN). Compare their expressive power, advantages and disadvantages.

4. **(Computation Graph, Full Batch)** (6p) Consider the undirected graph in the following figure, where node h is a hub connected to v and to six leaves $x_1, \ldots, x_6$.

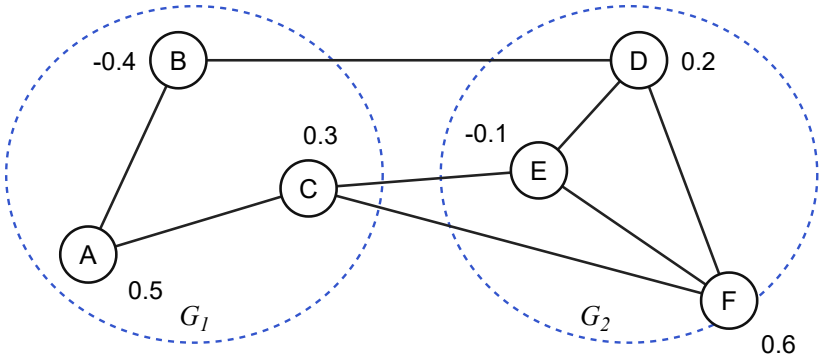


a. Draw the 2-layer computation graph of node v and count its nodes (a node is counted every time it appears in the tree). Do the same for node a. Compare both numbers with the number of nodes in the whole graph.

b. Explain the computational burden of computation graphs when more GNN layers are stacked. What is the role of the node degree distribution?

c. Why does GCN inherently have to be trained in full batch? In which situation does this become a problem?

5. **(Cluster-GCN)** (10p) Consider an undirected graph G of six nodes A, B, C, D, E and F given in the following figure. The graph G is partitioned into two clusters G_1 and G_2. Each node has an initial feature that is the number standing next to it. According to the Cluster-GCN model, the feature of a node i at layer k can be updated as:



$$h_{N(i)}^{(k)} = \text{MEAN}\big(\{h_u^{(k-1)},\ \forall u \in N(i),\ G_u = G_i\}\big)$$
$$h_i^{(k)} = \text{ReLU}\big(h_i^{(k-1)} \,\|\, h_{N(i)}^{(k)}\big)$$

where $\|$ is a concatenation.

a. Explain how Cluster-GCN builds the mini-batches used in SGD.

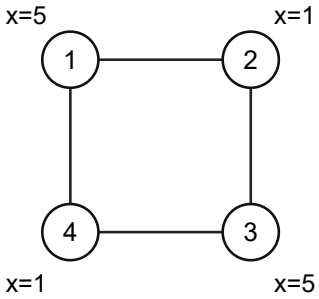
b. Calculate the output representations of all nodes at layer $k = 1$ (one cluster is one mini-batch).

c. Cluster-GCN greatly reduces the computational cost, but it also causes problems. Discuss these problems and how they can be mitigated.

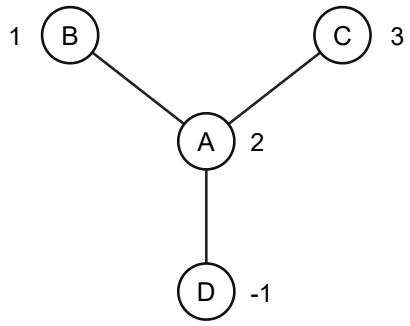
6. **(Deep GNNs, Low-pass Filter)** (8p)

a. Explain over-smoothing, over-squashing and under-reaching. What are they, and how do they show up as phenomena when GNNs are trained?

b. Consider the 4-cycle below with the node signal $x = [5,\ 1,\ 5,\ 1]^T$ and the GCN propagation matrix $P = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$. Since all nodes have the same degree, $P = \frac{1}{3}(A + I)$ here. Calculate Px and P^2x. Using the result, explain why averaging the neighbors' information acts as a low-pass filter.



7. **(GCNII)** (10p) Consider the undirected star graph in the following figure. Each node has a one-dimensional initial feature that is the number standing next to it. The output of a GCNII layer is calculated as:



$$H^{(\ell+1)} = \sigma\Big(\big((1-\alpha)\tilde{P}H^{(\ell)} + \alpha H^{(0)}\big)\big((1-\beta)I + \beta W^{(\ell)}\big)\Big)$$

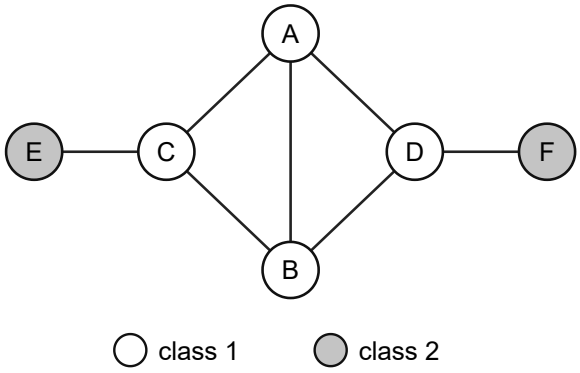
where $\tilde{P} = \tilde{D}^{-1/2}(A+I)\tilde{D}^{-1/2}$, $\tilde{D}$ is the degree matrix of $A+I$, $\alpha = 0.2$, $\beta = 0.5$, $W^{(0)} = [0.6]$, $I = [1]$ (the features are one-dimensional), and σ is a ReLU function.

a. What are the two major components introduced in GCNII? Point out which term of the formula implements each of them.

b. Calculate $\tilde{P}$ and the output representations $H^{(1)}$.

c. Explain how these two components mitigate the over-smoothing problem and allow GCNII to stack many more layers.

8. **(Homophily Metrics)** (10p) Consider the undirected graph in the following figure. Nodes A, B, C and D belong to class 1 and nodes E and F belong to class 2. The homophily metrics are defined as:



$$h_{edge} = \frac{|\{(u,v) \in E : y_u = y_v\}|}{|E|}$$

$$h_{node} = \frac{1}{|V|}\sum_{v \in V}\frac{|\{u \in N(v) : y_u = y_v\}|}{d_v}$$

$$h_{class} = \frac{1}{C}\sum_{k=1}^{C}\max\left(0,\ h_k - \frac{|V_k|}{|V|}\right), \qquad h_k = \frac{\sum_{v \in V_k}|\{u \in N(v) : y_u = k\}|}{\sum_{v \in V_k} d_v}$$

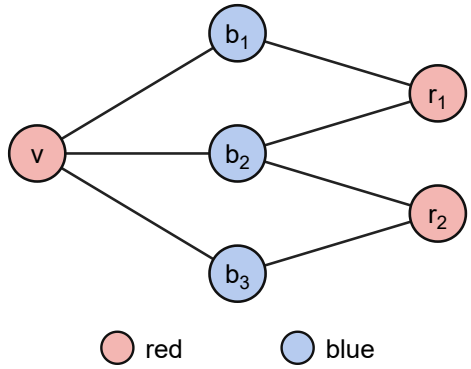
$$h_{adj} = \frac{h_{edge} - \sum_{k=1}^{C} p_k^2}{1 - \sum_{k=1}^{C} p_k^2}, \qquad p_k = \frac{\sum_{v \in V_k} d_v}{2|E|}$$

where V_k is the set of nodes of class k, C is the number of classes and d_v is the degree of node v.

a. Calculate h_{edge}, h_{node}, h_{class} and h_{adj} of this graph.

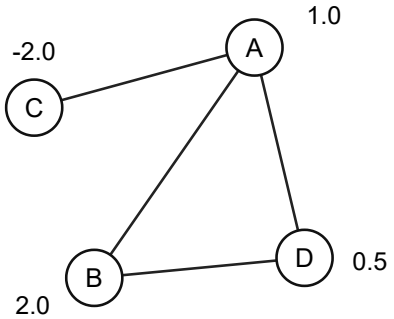
b. The four values are quite different. Explain what each metric measures and the differences in their meaning.

9. **(H2GCN)** (8p)
   a. Describe how H2GCN works, including its forward pass and its three key designs: (D1) separation of the ego embedding and the neighbor embedding, (D2) higher-order neighborhoods, and (D3) combination of intermediate representations.
   b. In the graph below, the node colors are the class labels. List the 1-hop neighbors $N_1(v)$ and the 2-hop neighbors $N_2(v)$ of node v, and give the fraction of nodes in each set that have the same label as v.



   c. Explain why each of the three designs helps GNNs under heterophily.

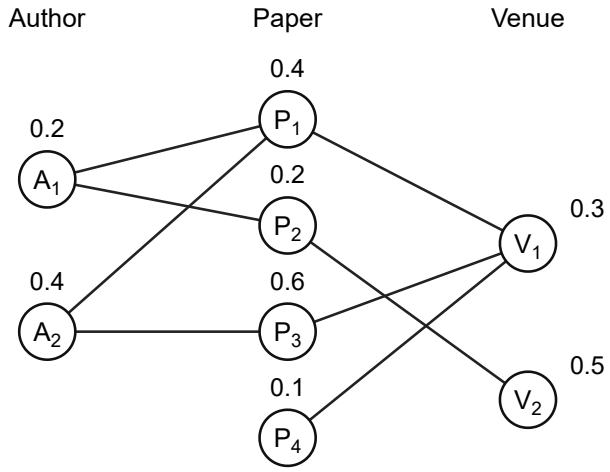
10. **(GAT, GATv2)** (10p) Consider an undirected graph of four nodes A, B, C and D given in the following figure. Each node has a one-dimensional initial feature that is the number standing next to it. In GAT, the weight matrix is $W = [0.5]$ and the attention vector is $a = [0.5,\ 0.5]^T$. The energy e_{ij}, the attention score α_{ij} and the output are:



$$e_{ij} = \mathrm{LeakyReLU}\big(a^T [Wh_i \,\|\, Wh_j]\big)$$
$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in N(i)} \exp(e_{ik})}$$
$$h_i^{(1)} = \sigma\Big(\sum_{j \in N(i)} \alpha_{ij} W h_j\Big)$$

   where $\mathrm{LeakyReLU}(x) = x$ if $x > 0$ and $0.02x$ otherwise, $\|$ is a concatenation, and σ is a ReLU function.
   a. Explain how the attention mechanism of GAT works.
   b. Calculate the energies e_{AB}, e_{AC}, e_{AD}, the attention scores α_{AB}, α_{AC}, α_{AD}, and the feature of node A at layer 1, $h_A^{(1)}$.
   c. GATv2 computes the energy as $e_{ij} = a_2^T\, \mathrm{LeakyReLU}\big(W_2 [h_i \,\|\, h_j]\big)$ with
$$W_2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad a_2 = [-1,\ -1]^T.$$
   Calculate the GATv2 energies e_{AB}, e_{AC}, e_{AD}. Which neighbor of A gets the largest attention in GAT (from b) and in GATv2?
   d. Explain how GAT and GATv2 differ and why they behave differently.

11. **(HAN)** (10p) Consider a heterogeneous graph given in the following figure. There are three types of nodes in the academic network: Author (A), Paper (P) and Venue (V). Each node has an initial feature that is the number standing next to it. According to HAN, every node is projected by $W = [0.5]$, i.e., $h'_i = W h_i$, and the node-level attention for a meta-path Φ is:



$$e_{ij}^{\Phi} = \mathrm{LeakyReLU}\big(a_{\Phi}^{T}[h'_i \,\|\, h'_j]\big), \quad a_{\Phi} = [0.5,\ 0.5]^{T}$$

$$\alpha_{ij}^{\Phi} = \frac{\exp(e_{ij}^{\Phi})}{\sum_{k \in N_i^{\Phi}} \exp(e_{ik}^{\Phi})}$$

$$z_i^{\Phi} = \sigma\Big(\sum_{j \in N_i^{\Phi}} \alpha_{ij}^{\Phi}\, h'_j \Big)$$

where N_i^{Φ} is the set of papers at the other end of the meta-path instances of Φ that start from paper i (excluding i itself), and σ is a ReLU function.

The semantic-level attention fuses the meta-paths with the semantic-level scores w_{Φ} as

$$\beta_{\Phi} = \frac{\exp(w_{\Phi})}{\sum_{\Phi'} \exp(w_{\Phi'})}, \qquad Z_i = \sum_{\Phi} \beta_{\Phi}\, z_i^{\Phi}.$$

a. List all the meta-path instances PAP and PVP that start from P_1. Calculate the node-level attention coefficients of P_1 for each meta-path and the meta-path specific embeddings $z_{P_1}^{PAP}$ and $z_{P_1}^{PVP}$.

b. Suppose the semantic-level scores (already averaged over all papers) are $w_{PAP} = 0.9$ and $w_{PVP} = 0.4$. Calculate β_{PAP}, β_{PVP} and the final embedding Z_{P_1}.

c. Explain how HAN works and why it uses attention twice (node-level and semantic-level).